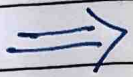


Practical No : 1

Test for proportion (one sample problem & two sample problem)

Q.1 A coin is tossed 400 times and it turned up head 210 times. Can the coin be regarded as fair? State the result using Confidence interval also use p-value criterion



P_0	0.5	Null Hypothesis $H_0 : P = P_0$
n	400	
x	210	
$q_0 = 1 - P_0$	0.5	Alternative hypothesis $H_1 : P \neq P_0$
$P(x/n)$	0.525	
q	0.475	
α	0.05	

Z-Statistics	$(P - P_0) / \text{SQRT}(P_0 q_0 / n)$ $= 1.959963985$
Z-Table	Two-tailed = Norm.S.Inv(1 - 2/2) $= 1.959963985$
$ Z < Z\text{-table}$	We accept null hypothesis

Conclusion :- The coin may be regarded as fair

p-value	Two-tailed test ($H_1 : P \neq P_0$) $= 2 * (1 - \text{Norm.S.DIST}(\text{ABS}(Z), \text{True}))$ $= 0.980054964$
---------	---

$P > 0.05 \rightarrow$ We accept null hypothesis

confidence interval

lower CI $\rightarrow P - \text{Norm.S.INV} (1 - \alpha/2) * \text{SQRT} (P * (1 - P/n))$

$$= 0.476062188$$

upper CI $\rightarrow P + \text{Norm.S.INV} (1 - \alpha/2) * \text{SQRT} (P * (1 - P/n))$

$$= 0.573937812$$

specified value (P_0) = 0.5 lie in confidence interval $\therefore$ we accept null hypothesis (H_0)

Q.2 In a sample of 500 parts manufactured by a company, the number of defective parts were found to be 42 company, however, claimed that 5% of their product is defective. is the claim valid? use 5% level of significance

level of significance

$\rightarrow$

P_0	0.06	
n	500	Null = $H_0: P = P_0$
x	42	
$P_0(1 - P_0)$	0.94	Alternative = $H_1: P \neq P_0$
$P(X/n)$	0.084	
q	0.916	
α	0.05	

Z-statistics = $(P - P_0) / \text{SQRT} (P_0 Q_0/n)$

$$= 2.259130731$$

Z-Table One tailed $\rightarrow = \text{Norm.S.INV} (1 - \alpha)$

$$= 1.95963985$$

|Z statistic| < |Z-table| $\therefore$ We accept null hypothesis

Conclusion: Product is defective is the claim valid.

P-value Two-tailed ($H_1: P \neq P_0$)

$$= 2 * (1 - \text{Norm.S.DIST} (ABS(Z), \text{True}))$$

$$= 0.023837968$$

Confidence interval

lower CI = $P - \text{Norm.S.INV} (1 - \alpha/2) * \text{SQRT} (P * (1 - P/n))$

$$= 0.059686$$

upper CI = $P + \text{Norm.S.INV} (1 - \alpha/2) * \text{SQRT} (P * (1 - P/n))$

$$= 0.10831367$$

Specified value (P_0) = 0.06 lie in confidence interval

$\therefore$ we accept null hypothesis (H_0)

Q.3 A manufacture claims that at least 85% of its batteries pass a quality check.

A safety inspector believes the true pass rate is lower than 0.85. A sample of $n = 200$ batteries is tested. Out of these

160 pass inspection. Test at $\alpha = 0.05$ whether the true pass rate is less than the claimed 85%.

$\Rightarrow$:

P_0	0.85	Null = $H_0: P = P_0$
n	200	
X	160	
$P_0 = 1 - P_0$	0.15	alternative = $H_1: P \neq P_0$
p	0.8	
q	0.85	
α	0.05	

Z-statistics $= (p - P_0) / \sqrt{P_0(1 - P_0)}$

$$= (0.8 - 0.85) / \sqrt{0.85 \times 0.15}$$

Z-Table $=$ left tailed $=$ Norm.S.Inv(α)

$$= -1.6448536$$

|Z-statistics| < Z-table | we reject Null hypo

Conclusion: The true pass rate is less than the claimed 85%.

P-value $\Rightarrow$ left tailed test ($H_1: P < P_0$)

$$= \text{Norm.S.Dist}(\alpha, \text{True})$$

$$= 1.952329619$$

$P < 0.05$ $\therefore$ We reject null hypothesis

X from were found to be smallest in a sample of 1000 persons from urban area 350 were

Q.4 Out of a sample of 100 residents in a certain area a 73 found to own their homes test the

hypothesis that proportion of house owners in 80% against the alternative that is less than 80%. Carry out the test using P-Value

$\Rightarrow$:

P_0	0.8	Null = $H_0: P = P_0$
n	100	
X	73	
$P_0 = 1 - P_0$	0.2	alternative = $H_1: P \neq P_0$
$P(X/n)$	0.73	
q	0.8	
α	0.05	

Z-statistic $= (p - P_0) / \sqrt{P_0(1 - P_0)}$

$$= -1.75$$

Z-Table left tailed $=$ Norm.S.Inv(α)

$$= -1.644853627$$

1/2 statistic < z-table we reject null hypothesis

Conclusion:- The alternative that is it less than 80%.

p-value $\Rightarrow$ left tailed test. (H1: $p < p_0$)
= Norm.S.dist (α . True)
= 1.91881686

Confidence interval

lower CI = $p - \text{norm.S.INV}(1 - \alpha/2) * \text{SQRT}(p * (1 - p/n))$
= 0.642985545
upper CI = $p + \text{norm.S.INV}(1 - \alpha/2) * \text{SQRT}(p * (1 - p/n))$
= 0.817014455

$p < 0.05$ we reject null hypothesis.

Q.5 In random sample of 800 persons from rural area 200 were found to be smokers. In a sample of 1000 persons from urban area 350 were found to be smokers. Test whether proportion of smokers is same for both populations. find 85% confidence interval for the diff of proportion smokers in rural & urban area

$\Rightarrow$

X1	200
x2	350
n1	800
n2	1000
α	0.05

$p = 1 - p = 0.694444$

Null hypothesis $\Rightarrow H_0: p_1 = p_2$
alternative $\Rightarrow H_1: p_1 \neq p_2$

z-statistic = $(p_1 - p_2) / \text{SQRT}(p * q * ((1/n_1) + (1/n_2)))$
= -4.576626248

z-Table Two tailed test = Norm.S.INV(C1 - $\alpha/2$)
= 1.959963985

1/2 statistic < z table we accept Null hypothesis

Conclusion:- The proportion of smokers in rural area is not same as proportion of smoker in rural area

p-value $\Rightarrow$ Two tailed test (H1: $p_1 \neq p_0$)
= 2 * (1 - Norm.S.DIST(ABS(z), True))
= 4.72548

$p < 0.05$ $\therefore$ we reject null hypothesis

Confidence Interval

$$- \text{lower CI} = (p_1 + p_2) - \text{Norm.S.INV}(1 - \alpha/2) * \sqrt{p * q * (1/n_1 + 1/n_2)}$$

$$= -0.142825574$$

$$- \text{upper CI} = (p_1 + p_2) + \text{Norm.S.INV}(1 - \alpha/2) * \sqrt{p * q * (1/n_1 + 1/n_2)}$$

$$= -0.057174426$$

0 value does not lie in confidence interval
 $\therefore$ we reject H_0 .

Q.6 A certain factory, run in two shift. A

Sample of 1000 items Selected from production of day shift, gave 52 defective articles. however a sample of 700 items selected from production of night shift revealed 45 items defective. Can we conclude that proportion of defective items shifts is less than of second shift.

$\Rightarrow 7\%$

X_1	52
X_2	42
n_1	1000
n_2	700
α	0.05

$$p_1 = (X_1/n_1) = 0.052$$

$$p_2 = (X_2/n_2) = 0.06$$

$$p = \frac{(X_1 + X_2)}{(n_1 + n_2)} = 0.055294115$$

$$\Phi = 1 - p = 0.94470582$$

$$z\text{-Statistic} = (p_1 - p_2) / \sqrt{p * q * (1/n_1 + 1/n_2)}$$

$$= -0.710275395$$

Z-Table

$$\text{Two tailed} = \text{Norm.S.INV}(1 - \alpha/2) = 1.959663985$$

|Z Statistic| < Z-table $\therefore$ we accept H_0

Conclusion: The proportion of defective items in the first shift is less than of second shift.

P-value $\Rightarrow$ Two tailed test ($H_1: p \neq p_0$)

$$= 2 * (1 - \text{Norm.S.DIST}(\text{ABS}(z), \text{True}))$$

$$= 1.52246625$$

$p < 0.05$ $\therefore$ we reject null hypothesis

Confidence interval

lower CI = $(\rho_1 - \rho_2) - \text{Norm.S.INV}(1 - \alpha/2) * \text{SQRT}(\rho * q * (1/n_1) + (1/n_2))$

= -0.022165644

upper CI = $(\rho_1 - \rho_2) + \text{Norm.S.INV}(1 - \alpha/2) * \text{SQRT}(\rho * q * (1/n_1) + (1/n_2))$

= 0.021985433

Large Sample Test for Mean
Practical No: 2

Q.1) An automatic machine is designed to fill tins with 2 kilogram of oil with a standard deviation of 0.1 kg. a sample of 100 tins was examined and the average weight was found to be 1.94 kg. can we say that the machine is working properly? also construct 95% confidence interval of mean weight of tins

X	1.94	Null : $H_0: \mu = \mu_0 = 2$
μ_0	2	
S.d	0.1	Alternative $H_1: \mu \neq \mu_0 \neq 2$
n	100	
α	0.05	

Z statistic	$= (X - \mu_0) / (S.d / \text{SQRT}(n))$
	$= -6 \quad -6 $
	$= 6$
Z-Table	$= \text{Norm.S.INV}(1 - \alpha/2)$
	$= 1.959963985$

Z statistic | Z table $\therefore$ We reject null hypothesis

Conclusion: Machine does not work properly
p-value $\Rightarrow = 2 * (1 - \text{Norm.S.DIST}(\text{ABS}(Z)))$
 $= 1.97318 \text{E-}09$

$p < 0.05 \therefore$ We reject null hypothesis

Confidence interval

$$\text{lower CI} = \bar{X} - \text{Norm.S.INV}(1-\alpha/2) * (S.d) / \sqrt{n}$$

$$= 1.92040036$$

$$\text{upper CI} = \bar{X} + \text{Norm.S.INV}(1-\alpha/2) * (S.d) / \sqrt{n}$$

$$= 1.93959964$$

(1.92640036, 1.93959964)
specified value (μ_0) = 2 does not lie in confidence interval. we reject H_0 .

Q.2

The mean diastolic blood pressure for a group of 81 adults was found to be 79.2 mm. Test the hypothesis that the mean diastolic blood pressure is 75 mm. Population standard deviation is known to be 9 mm. also find confidence interval for mean diastolic blood pressure.

 $\Rightarrow$

X	79.2	Null	$H_0: \mu = \mu_0 = 75$
μ_0	75		
S.d	9	alternative	$\mu \neq \mu_0 \neq 75$
n	81		
d	6.05		

Z-Statistic

$$= (\text{mean} - \mu_0) / (S.d / \sqrt{n})$$

$$= 4.2$$

Z-Table

$$= \text{Norm.S.INV}(1-\alpha/2)$$

$$= 1.939964$$

1/2 Statistic < Z table $\therefore$ we reject null hypothesis

Conclusion:- Specified value = 75 not lie in confidence interval. we reject H_0

$$\text{p-value} = 2 * (1 - \text{Norm.S.DIST}(ABS(z)))$$

$$= 2.66919E-05$$

$p < 0.05$ $\therefore$ We reject H_0

Confidence interval

$$\text{lower CI} = \bar{X} - \text{Norm.S.INV}(1-\alpha/2) * (S.d) / \sqrt{n}$$

$$= 77.24004$$

$$\text{upper CI} = \bar{X} + \text{Norm.S.INV}(1-\alpha/2) * (S.d) / \sqrt{n}$$

$$= 81.15996$$

Q.3)

A sample of 900 ball bearings is found to have average weight of 12.5 grams. Can we assume that a sample is coming from a population with mean 13 grams again that it is than 13 grams? assume that the populations standard deviation is 1 gram. also construct 95% confidence interval

⇒

X	125	Null Hypothesis: $\mu_0 = 13$
μ_0	13	
S.d	1	Alternative H1: $\mu \neq \mu_0 \neq 13$
n	900	5% level of significance
α	0.05	

Z-statistic = $(\bar{X} - \mu_0) / (S.d / \sqrt{n})$

= -15 (H₀)

= 15 (population standard deviation)

Z-Table = Norm.S.INV(α)

= -1.64853627

|Z-statistic| > Z-table ∴ we reject Null (H_0)

Conclusion ∴ A Sample may not coming from population with mean 13

P-value ⇒ = Norm.S.DIST(Z)

= 3.67097E-5

$P < 0.05$ ∴ we reject H_0

confidence interval

Lower CI = $\bar{X} - \text{Norm.S.INV}(1-\alpha) * (S.d)$

SQRT(n)

= 1243466787

Upper CI = $\bar{X} + \text{Norm.S.INV}(1-\alpha) * (S.d)$

SQRT(n)

= 1256533213

Specified value (μ_0) = 13 does not lie in confidence interval we reject H_0

Q.4

A machine is supposed to fill bottles with 5.00 liters of oil. The population standard deviation is 0.20 L. A random sample of 64 bottles gave an average fill of 5.06 L. Test whether the machine overfills the bottles at the 5% level of significance

⇒

X	5.06
μ_0	5.00
S.d	0.20
n	64
α	0.05

Q.3)

A sample of 900 ball bearings is found to have average weight of 12.5 grams. Can we assume that a sample is coming from a population with mean 13 grams again that it is than 13 grams? assume that the populations standard deviation is 1 gram. also construct 95% confidence interval

 $\Rightarrow$

X	125	Null Hypothesis: $\mu_0 = 13$
μ_0	13	
s.d	1	Alternative $H_1: \mu \neq \mu_0 \neq 13$
n	900	
α	0.05	

Z-statistic

$$= (X - \mu_0) / (s.d / \sqrt{n})$$

$$= -15 \text{ HSI}$$

$$= 15 \text{ (population specification)}$$

Z-Table

$$= \text{Norm.S.INV}(\alpha)$$

$$= -1.64853627$$

Z-statistic > Z table $\therefore$ we reject Null (H_0)

Conclusion $\therefore$ A sample may not coming from population with mean 13

$$P\text{-value} \Rightarrow = \text{Norm.S.DIST}(Z)$$

$$= 3.67097E-51$$

$P < 0.05 \therefore$ we reject H_0

confidence interval

$$\text{Lower CI} = X - \text{Norm.S.INV}(1-\alpha) * (s.d)$$

$$= 12434.66787$$

$$\text{Upper CI} = X + \text{Norm.S.INV}(1-\alpha) * (s.d)$$

$$\text{SQRT}(n)$$

$$= 1256533213$$

Specified value (μ_0) = 13 does not lie in confidence interval we reject H_0 .

Q.4

A machine is supposed to fill bottles with 5.00 liters of oil. The population standard deviation is 0.20 L. A random sample of 64 bottles gave an average fill of 5.06 L. Test whether the machine overfills the bottles at the 5% level of significance

 $\Rightarrow$

X	5.06
μ_0	5.00
s.d	0.20
n	64
α	0.05

Z-Statistic $= (X - \mu_0) / (S.d / \sqrt{n})$

$= 2.4$

Z-Table $= \text{Norm.S.INV}(\alpha)$

$= -1.64485$

$= 1.64485$

|Z Statistic| > Z table $\therefore$ we reject H_0

Conclusion:- The machine overfills bottles at the 5% level of significance

P-value $\Rightarrow$ $= 1 - \text{Norm.S.DIST}(Z, \text{True})$
 $= 0.016365$

$P < 0.05$ $\therefore$ we reject H_0 and conclude A

Confidence interval

Lower CI $\Rightarrow X - \text{Norm.S.INV}(1 - \alpha/2) * S.d$
 $\sqrt{n}$
 $= 5.618879$

Upper CI $\Rightarrow X + \text{Norm.S.INV}(1 - \alpha/2) * (S.d) / \sqrt{n}$
 $= 5.101121$

The 95% confidence interval for the average fill is 5.10, 5.62 liters.

Q.5 A sample of 400 electric bulbs from company A gave an average life 1225 hours, with a standard deviation 42 hours, where as sample of 200 bulb from company B gave an average life 1265 hours with a S.D 60 hours. Can we say that the two companies are producing bulbs of same average life.

$\bar{X}_1$	1225	$\mu_1 = \mu_2$
$\bar{X}_2$	1265	
$S.d(X_1)$	42	
$S.d(X_2)$	60	Alternative $H_1: \mu_1 \neq \mu_2$
n_1	400	
n_2	200	

Z-statistic $= (\bar{X}_1 - \bar{X}_2) / \sqrt{S.d(X_1)^2 / n_1 + S.d(X_2)^2 / n_2}$
 $= -8.44965$

$= 8.44965$

Z-Table $= \text{Norm.S.INV}(1 - \alpha/2)$
 $= 1.959963$

|Z Statistic| > Z table $\therefore$ we reject H_0
Conclusion:- two companies not produce bulb of same average life.

P-value $= 2 * (1 - \text{Norm.S.DIST}(|Z|, \text{True}))$
 $= 0$

P value < 0.05 $\therefore$ we reject H_0

Confidence interval

$$\pm \text{lower CI} = (\bar{X}_1 - \bar{X}_2) - \text{Norm.S.INV}(1 - \alpha/2) * \text{SQRT}((\sigma_1^2/n_1) + (\sigma_2^2/n_2))$$

$$= -49.278313$$

$$\pm \text{upper CI} = (\bar{X}_1 - \bar{X}_2) + \text{Norm.S.INV}(1 - \alpha/2) * \text{SQRT}((\sigma_1^2/n_1) + (\sigma_2^2/n_2))$$

$$= -36.721687$$

Q.6 A random sample of 400 persons from country A gave mean height 170 cm.

another sample of person's from country B are taller than those of A given that their product population S.D are given respectively.

A	B	S
$\bar{X}_1 = 170$	$\bar{X}_2 = 180$	
$n_1 = 400$	$n_2 = 800$	
$\sigma_1 = 8$	$\sigma_2 = 8$	
$\alpha = 0.05$	$\alpha = 0.05$	

Z-statistic

$$= (\bar{X}_1 - \bar{X}_2) / \text{SQRT}((\sigma_1^2/n_1) + (\sigma_2^2/n_2))$$

$$= 19.4033 = -26.5867$$

Z-table

$$= \text{Norm.S.INV}(\alpha) = 1.64$$

12 statistic < Z table $\therefore$ we reject H_0

Conclusion: 'B' are not taller than A

$$P\text{-value} \Rightarrow = 2 * (1 - \text{Norm.S.DIST}(\text{ABS}(Z), \text{True}))$$

$$= 2$$

P value < 0.05 $\therefore$ we reject H_0

Q.7)

In a college there are two faculties Arts and Science. The average weight of students in a sample of 250 in arts faculty was found to be 120 lbs. with standard deviation of 12 lbs. while the corresponding fig in a sample of 400 students from science faculty were 124 lbs is this diff significant?

$$\Rightarrow \frac{x_1}{n_1} = \frac{120}{250} \quad \frac{x_2}{n_2} = \frac{212}{400}$$

$$n_1 = 250$$

$$n_2 = 400$$

$$\sigma_1 = 12$$

$$\sigma_2 = 14$$

$$d = 0.05$$

$$d = 0.05$$

$$H_0: \mu_1 = \mu_2 \quad \text{Null}$$

$$H_1: \mu_1 \neq \mu_2 \quad \text{Alternative}$$

2-statistic

$$= (x_1 - x_2) \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

$$= -3.87419$$

P-table

$$= 2 * (1 - \text{Norm.S.DIST}(2, \text{True}))$$

$$= 1.999893$$

2-table

$$= \text{Norm.S.INV}(1 - \alpha/2)$$

$$= 1.98964$$

P-value < 0.05, we reject H_0

~~2-table~~

Practical No. 3

Test for means (one sample problem) (small sample set)

Q.1

A company claims that the average diameter of metal rods is 10 mm. A sample of 10 rods given a mean diameter of 10.4 mm. The population S.D is known to be 1.2 mm. Test at 5% significance level whether the rods differ in mean diameter from the claimed value.

X	10.4
n	12
μ_0	10
α	0.05
S.d	1.2

$$\text{Null: } H_0: \mu = \mu_0$$

$$\text{Alternative: } H_1: \mu \neq \mu_0$$

z-statistic

$$= (x - \mu_0) / (S.d / \sqrt{n})$$

$$= 1.154700538$$

Table value

$$= \text{Norm.S.INV}(1 - \alpha/2)$$

$$= 1.98963$$

1/2 statistic < table value

we accept null hypothesis (H_0)

P-value $\Rightarrow$ Two tailed test ($H_1: \mu \neq \mu_0$)

$$= 2 * (1 - \text{Norm.S.DIST}(\text{ABS}(z), \text{True}))$$

$$= 0.248213079$$

$P > 0.05$: we accept null hypothesis

confidence interval
 $\text{lower CI} = (X) - \text{Norm.S.INV}(1 - \alpha/2) * (S.d) / \text{SQRT}(n)$

$\text{upper CI} = (X) + \text{Norm.S.INV}(1 - \alpha/2) * (S.d) / \text{SQRT}(n)$

$$= 9.72104856$$

$$\text{upper CI} = (X) + \text{Norm.S.INV}(1 - \alpha/2) * (S.d) / \text{SQRT}(n)$$

$$= 11.67895144$$

Q.2) A food manufacturer states that the mean weight of a snack packet is 50 g. A sample of 15 packs shows a mean weight of 48.8 g. The population standard deviation is known to be 2.5 g. Test at 5% level whether the packets are underweight.

$\Rightarrow$

X	48.8	$H_0: \mu = 50$
n	15	
μ_0	50	$H_1: \mu \neq 50$
α	0.05	
S.d	2.5	

Z-statistic = $(X - \mu_0) / (S.d / \text{SQRT}(n))$

$$= -1.794432438$$

$$= 1.794432438$$

table value = Norm.S.INV(α)

$$= -2.3263478$$

$$= 2.3263478$$

1/2 statistic < table value

we accept null hypothesis (H_0)

P-value $\Rightarrow$ one tailed test ($H_1: \mu \neq \mu_0$)

$$= 2 * (1 - \text{Norm.S.DIST}(\text{ABS}(Z), \text{True}))$$

$$= 0.672744123$$

$P > 0.05 \rightarrow$ we accept H_0 .

confidence interval

lower CI = $(X) - \text{Norm.S.INV}(\alpha/2) * (S.d) / \text{SQRT}(n)$

$$= 50.35571054$$

upper CI = $(X) + \text{Norm.S.INV}(\alpha/2) * (S.d) / \text{SQRT}(n)$

$$= 47.24428946$$

Q.3 The average time taken to solve a puzzle is claimed to be 30 minutes. A sample of 20 participants solves it in an average of 28.5 minutes. The population S.D is known to be 4 minutes. Test at 5% level whether the mean time differs from the claimed average.

⇒

X	28.5	$H_0: \mu = \mu_0$
n	20	
μ_0	30	$H_1: \mu \neq \mu_0$
σ	0.05	
S.d	4	

Z-Statistic = $(X - \mu_0) / (S.d / \sqrt{n})$

$$= -1.677650983$$

$$= 1.677650983$$

Table value = Norm.S.INV(1-d/2)

$$= 1.959963$$

|Z-Statistic| < Table value

∴ we accept H_0 (Null)

p-value ⇒ Two tailed test $(H_1: \mu \neq \mu_0)$

$$= 2 * (1 - \text{Norm.S.DIST}(\text{ABS}(Z), \text{True}))$$

$$= 0.693532513$$

p > 0.05 ∴ we accept H_0

confidence interval

Lower CI = $(X) - \text{Norm.S.INV}(1-\alpha/2) * (S.d)$

$$\sqrt{n}$$

$$= 26.74695492$$

Upper CI = $(X) + \text{Norm.S.INV}(1-\alpha/2) * (S.d) / \sqrt{n}$

$$= 30.25304508$$

(Q.4) A chemist claims that the average pH value of a solution is 7.0. The following pH readings are recorded: 6.8, 7.2, 7.1, 6.9, 7.3, 7.0, 6.7.

Test at the 5% significance level whether the mean pH differs from 7.0

X	7	(x_i)	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
n	7	6.8	-0.2	0.04
μ_0	7	7.2	0.2	0.04
σ	0.5	7.1	0.1	0.01
		6.9	-0.1	0.01
		7.3	0.3	0.09
		7	0	0
		6.7	0.3	0.09
		Total	49	0.28

~~$H_0: \mu = \mu_0$~~

$H_1: \mu \neq \mu_0$

$$S = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$$

$$= 0.68192$$

$$T \Rightarrow (X - \mu_0) / (S / \sqrt{SPRT(n)}) \cdot \sqrt{(n-1)}$$

$$= 0$$

$$df \Rightarrow T - 1 = 27 - 1 = 26$$

$$\text{Critical value} = T.INV.2T(\alpha, df)$$

$$= 2.446911851$$

$T < \text{critical value}$

we accept H_0

$$P\text{-value} \Rightarrow 2 * (1 - T.DIST.2T(ABS(T), n-1))$$

$$= 0$$

~~confidence interval~~

$$\text{lower CI} = X - T.DIST.2T(\alpha, n-1) * (S / \sqrt{SPRT(n)})$$

$$= 6.967941823$$

$$\text{upper CI} = X + T.DIST.2T(\alpha, n-1) * (S / \sqrt{SPRT(n)})$$

$$(n)$$

$$= 7.032058177$$

Q.5 A company says the average lifespan of a small electronic component is 450 hours.

A sample from production gives the following lifetimes (in hours): 445, 452, 460, 448,

455, 462, 449, 453. Test at the 5%.

Significance level whether the data supports the company's claim.

X	453	$H_0: \mu = \mu_0$
h	8	
μ_0	450	$H_1: \mu \neq \mu_0$
α	0.05	

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$	$S = \sqrt{SPRT(240) / (n-1)}$
445	-8	64	$= 2.213133341$
452	-1	1	
460	7	49	$T \neq (X - \mu_0) / (S / \sqrt{SPRT(n)})$
448	-5	25	
455	2	4	$= 3.384077093$
462	9	81	
449	-4	16	$df = (n-1)$
453	0	0	
Total	-3624	0	$= 7$

$$\text{Critical value} \Rightarrow T.INV.2T(\alpha, df)$$

$$= 2.364624252$$

$T < \text{critical value}$

we reject null hypothesis

$$P\text{-value} \Rightarrow 2 * (1 - T.DIST.2T(ABS(T), n-1))$$

$$(T, n-1)$$

$$= 1.987152252$$

Confidence interval

$$\text{lower CI} = X - T \cdot \text{DIST} \cdot 2T (\alpha, n-1) * (S / \text{SQRT}(n))$$

$$= 4522476489$$

$$\text{upper CI} = X + T \cdot \text{DIST} \cdot 2T (\alpha, n-1) * (S / \text{SQRT}(n))$$

$$= 453352311$$

Q.6 A researcher believes that the mean heart rate of a certain species of birds is 120 beats per minute. The observed heart rates are: 118, 122, 119, 121, 124, 117, 123. Test at the 1% significance level whether the mean heart rates differs from 120 bpm

→

$\bar{X}$ (mean)	120.57	$H_0: \mu = \mu_0$
n	7	$H_1: \mu \neq \mu_0$
μ_0	120	
α	0.01	

X_i	$(X_i - \bar{X})$	$(X_i - \bar{X})^2$
118	-2.5	6.6122
112	1.4	2.040
119	-1.5	2.469
121	0.4	0.183
124	8.4	11.781
117	3.5	12.75
123	2.4	5.89

Total

844	1.2410	41.71
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$$S = \text{SQRT} (441.71) / (n-1)$$

$$= 1.07644329$$

$$T = (X - \mu_0) / (S / \text{SQRT}(n))$$

$$= 1.404493$$

$$df = (n-1)$$

$$= 6$$

$$\text{Critical} = T \cdot \text{INV} \cdot 2T (\alpha, df)$$

$$= 3.707428$$

$T < \text{Critical Value}$
we accept null hypothesis

$$P\text{-value} \Rightarrow = 2 * (1 - T \cdot \text{DIST} \cdot 2T (\text{ABS}(T), n-1))$$

$$= 1.580468114$$

Confidence interval

$$\text{lower CI} = X - T \cdot \text{DIST} \cdot 2T (\alpha, n-1) * (S / \text{SQRT}(n))$$

$$= 120.97$$

$$\text{upper CI} = 120.97$$

Practical No: 4

Test for means (two sample problem)

- Q.1) The gain in weights (in lbs) of pigs fed on two diets A and B given below. Test whether the two diets differ significantly regarding their effect on increase in weight.

Diet	gain in weight
A	25, 32, 30, 34, 24, 14, 32, 24, 30, 31, 35, 25
B	44, 34, 42, 10, 47, 31, 40, 30, 32, 35, 18, 21, 18, 35

⇒ Q.2 - T-test - two-sample assuming equal variances

	Variable 1	Variable 2
mean	28	30
variance	34.5454	100.71
observation	12	15
pooled variance	71.6	
hypothesized mean diff	0	
df	25	
t stat	-0.61027	
p (T ≤ t) one tail	0.273392	
t critical one-tail	1.708146	
p (T ≤ t) two tail	0.5471856	
t critical two tail	1.21059385	

H₀: $\mu_1 = \mu_2$
H₁: $\mu_1 \neq \mu_2$

T stat < T critical - we fail to reject H₀p > 0.05 - we fail reject H₀

Conclusion: There is no statistically significant diff in mean gain b/w pigs fed on diet A and B at the 5% significance.

The test scores of two teaching methods A and B are given below. Test whether the two methods differs significantly.

method A	56, 60, 65, 58, 70, 62, 68, 64
method B	72, 75, 70, 68, 74, 80, 77, 69, 73

⇒ Q.3 - T-test: Two-sample assuming equal variance

	Variable 1	Variable 2
Mean	62.875	73.1111
variance	23.26785	18.1111
observation	8	9
pooled variance	18.91759	
hypothesized mean diff	0	
df	15	
t-stat	-4.843327	
p (T ≤ t) one tail	0.0001074	
t critical one-tail	1.753050356	
p (T ≤ t) two tail	0.00021	
t critical two tail	2.13144	

Q.3) Two fertilizers F_1 and F_2 were tested on crop yield (in kg). examine whether fertilizer F_2 produces a significantly higher yield than fertilizer F_1 .

fertilizer F_1	110, 95, 105, 120, 100, 115, 108
fertilizer F_2	136, 125, 118, 140, 135, 128, 132

t-test: Two-Sample assuming equal variance

	variable 1	variable 2
mean	107.57142	129.71
variance	72.95	80.2380
observation	7	7
pooled variance	61.5952	
hypothesized mean diff	0	
df	12	
t-stat	-5.2783005	
$P(T \leq t)$ one tail	9.73703	
t critical one tail	1.78228	
$P(T \leq t)$ two tail	0.000195	

$H_0: \mu_1 = \mu_2$
 $H_1: \mu_1 \neq \mu_2$
 $T \text{ Stat} > T \text{ Critical} \therefore \text{we reject } \mu_1$

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$P < 0.05 \therefore \text{we reject } H_0$

Conclusion:- fertilizer F_2 produces a significantly higher yield than fertilizer F_1 .

Q.4) The reaction times (in seconds) for two groups of drivers using diff training program X and Y are:

program X	1.2, 1.5, 1.3, 1.6, 1.4, 1.7
program Y	1.8, 2.0, 1.7, 2.1, 1.9, 2.2, 1.8

t-test: Two sample assuming equal variance

	variable 1	variable 2
mean	1.45	1.9285
variance	0.035	0.0323
observation	6	7
pooled variance	0.0335714	
hypothesized mean diff	0	
df	11	
t-stat	-4.69477	
$P(T \leq t)$ one tail	0.00032	
t critical one tail	1.79588	
$P(T \leq t)$ two tail	0.00065	
t critical two tail	2.20098	

$H_0: \mu_1 = \mu_2$
 $H_1: \mu_1 \neq \mu_2$

$T_{stat} > T_{critical}$ $\therefore$ we reject null hypothesis

$p < 0.05$ $\therefore$ we reject null hypothesis

Conclusion $\therefore$ This is statistically significant
diff in two group driver in program X and
Y at the 5% significance

Practical No: 5

Paired-t-test

The hours of sleep for 10 patients before and after giving a new drug and recorded. Test whether there is a significant diff in the average average hours of sleep at 5% level of significance.

Patient no	1	2	3	4	5	6	7	8	9	10
Hours of (before)	6	5	7	7	8	9	6	6	7	8
Hours of (after)	7	8	7.5	9	7	6.5	7	8	8.5	7

T-test: Paired two sample for means

	Variable 1	Variable 2
Mean	6.9	7.53
variance	1.43333	0.63611
observation	10	10
pearson correlation	-0.4015	
hypothesized mean diff	0	
df	9	
t-stat	-1.22053	
$P(T \leq T)$ one tail	0.126636	
t critical one tail	1.8331129	
$P(T \leq t)$ two tail	0.253273	
t critical two tail	2.262157	

$H_0: \mu d = 0$

$H_1: \mu d \neq 0$

$T_{stat} < T_{critical}$ $\therefore$ we reject null (H_0)

$p > 0.05$ $\therefore$ we reject H_0

Conclusion:- there is no statistically significant difference in average hours of sleep before and after giving a new drug at after 5% significance

Q.2 The hours of sleep (or measurement) for 10 individual were recorded before and after a treatment or intervention a given below

before	78	82	80	76	84	79	81	77	83	80
after	72	75	74	70	78	73	74	71	76	85

Test, at the 5% level of significance, whether there is a significant difference in the measurement before and after the treatment.

$\Rightarrow$ T-test: paired two sample for means

	Variable 1	Variable 2
Mean	80	73.8
Variance	6.66666	15.73333
Observation	10	10
PC	6.97049	
HMD	6	
df	9	
Tstat	31	

$p (t < T)$ one tail 9.26879

t critical one tail 1.8331129

$p (t < T)$ two tail 1.85376

t critical two-tail 2.262157

$H_0: \mu_d = 0$

$H_a: \mu_d \neq 0$

$T_{stat} > T_{critical}$ $\therefore$ we reject H_0

$p > 0.05$ $\therefore$ we reject H_0

Conclusion:- there is no statistically significant difference in the sleep hours of before and after a treatment after at the 5% significance.

Q.3 A doctor tests a new drug on 8 patients. measures systolic blood pressure before and after treatment. Test whether the drug significantly reduces BP.

Patient	1	2	3	4	5	6	7	8
before	150	160	155	148	162	158	152	149
after	140	148	145	138	150	147	142	139

$\Rightarrow$ T-test: Paired two sample for means

	Variable 1	variable 2
mean	154.25	143.62
Variance	28.214285	20.26716
observation	8	8
pearson correlation	0.996160	
h.M.D	0	
df	7	
t stat	32.80311	
$P(T \leq t)$ one tail	3.16629	
t critical value	1.8945786	
$P(T \leq t)$ two tail	6.33258	
t critical two tail	2.364624	

$$H_0: \mu_d = 0$$

$$H_1: \mu_d \neq 0$$

$$T_{stat} > T_{critical} \therefore \text{we reject } H_0$$

$$P > 0.05 \therefore \text{we reject } (H_0)$$

Conclusion: there is no statistically significant

diff in the new drug of measure systolic

bp before a treatment after at the 5% level

Q.4 A fitness trainer records the weights (kg) of 9 clients before and after a 6-week diet program. Verify if the program significantly reduces weights

Before	82	75	90	85	92	78	88	80	84
after	89	72	86	81	88	75	84	76	81

T-test: Paired two sample for means

	Variable 1	Variable 2
Mean	83.7777	80.2222
Variance	31.69444	27.9444
observation	9	9
pearson corr	0.997315	
H.M.D	0	
df	8	
stat-t	26.23857	
$P(T \leq t)$ one tail	1.88867	
t critical one tail	1.859548	
$P(T \leq t)$ two tail	3.71135	
t critical two tail	2.3060641	

Chi-square test for independence of attributes

Practical No: 6

A.1 A study was conducted to determine if there is an association between gender and preference for a type of snack (Chips & Candy)

The data is as follows:

Gender	Chips	Candy	Total
Male	30	20	50
Female	40	60	100
Total	70	80	150

perform a chi-square test for independence at the 0.05 significance level.

- 1) state the null and alternative hypothesis.
- 2) Calculate the expected freqⁿ for each cell
- 3) calculate the chi-square statistic.
- 4) determine the p-value and interpret the result.

Ans:

oij				N = 150
Gender	Chips	Candy	Total	
male	30	20	50	
female	40	60	100	
Total	70	80	150	

Expected Frequency (EIJ)

Gender	Chips	Candy	Total
male	$= (70 \times 50) / 150$	$= (80 \times 50) / 150$	50
female	$= (70 \times 100) / 150$	$= (80 \times 100) / 150$	100
Total	70	80	150

gender	Chips	Candy	Total
male	23.3333	26.6666	50
female	46.6666	53.3333	100
Total	70	80	150

$$OIJ^2 / EIJ = 30^2 / 23.333$$

gender	Chips	Candy	Total
male	38.571428	15	53.5714
female	34.2857	67.5	101.786
Total	72.8571	82.5	155.357

H₀: the two variable are independent

H₁: The two variable are not independent

$$\chi^2 \text{ statistic } (X) = \text{Total of } OIJ^2 / EIJ - N$$

$$= 155.357 - 150 = 5.35714$$

$$df = (R - 1) * (C - 1)$$

$$= 2 - 1 * 2 - 1 = 1 * 1 = 1$$

$$\text{Table value} \Rightarrow \chi^2_{(1, 0.05, df)} = 3.84145$$

Since $\chi^2 \text{ statistic} < \text{Table value}$ we reject H₀

$$p\text{-value} = \chi^2_{\text{DIST}}(t, df, \text{True}) = 0.97936$$

$p < 0.05$ $\therefore$ we reject H_0

Conclusion:- There is a associated betⁿ gender & preference for a type of snack (Chips & Condy)

Q.2) consider the following contingency table showing the preference for a new product betⁿ two regions (North and South). The test for independence was conducted, and the calculated chi-square statistic is 6.4 with 8 degrees of freedom.

1) What is the p-value for this test?

2) At a significance level of 0.05, what conclusion

Can you draw regarding the relationship betⁿ regions and product preference?

OIT

Region	Yes	No	Total
North	50	30	80
South	40	60	100
Total	90	90	180

(row sum + column sum) / N = (180 - 90) / 180 = 0.5

exp freq (EIT)

Region	Yes	No	Total
North	40	40	80
South	50	50	100
Total	90	90	180

OIT ² / EIT	Yes	No	Total
Region	62.5	22.5	85
North	32	72	104
South	94.5	94.5	189
Total	157	117	274

$$\chi^2\text{-stat} (x) = \frac{\text{total of OIT}^2 / \text{EIT} - N}{N}$$

$$= 9$$

$$df = (2-1) * (2-1)$$

$$= 1$$

$$\text{table value} = \chi^2_{(1, 0.05, 1)} = 3.841458$$

$$12 \text{ stat} < \text{table value} \therefore \text{we reject } H_0$$

$$p\text{-value} = \chi^2_{(1, 0.05, 1)} (X, df, \text{True})$$

$$= 0.907300204$$

$p < 0.05$ $\therefore$ we reject H_0

Conclusion:- fail to reject null hypothesis
there is no signi betⁿ region & product

Q.3)

A Survey was conducted to study whether customer preference for online or in-store shopping is independent of age group.

The data is summarized in the following table.

Suppose u and v are independent if you follow normal and v follow χ^2 distribution with n of freedom

then $z = u / \sqrt{v/n}$ is a test

classmate

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	online	In-Store	Total
18-25 years	40	60	100
26-40 years	50	70	120
41+ years	30	50	80
Total	120	180	300